

AI Assignment 1

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∴ Water Jug Problem ∴

we have two jugs

- i) 4-litre jug
- ii) 3-litre jug.

Goal ∴

exactly 4-litre water in
the 4-litre jug.

States ∴

= A state is represented as;
 (x, y) .

x = water in 4-litre jug

y = water in 3-litre jug.

actions ∴

$(0, y)$	empty 4-litre jug
$(x, 0)$	empty 3-litre jug
$(4, y)$	Fill 4-litre
$(x, 3)$	Fill 3 litre jug.
$(x-d, y+d)$	pour from 4-litre to 3lit
$(x+d, y-d)$	pour from 3litre to 4.

Transition Model:

- $(x, y) \rightarrow (4, y)$
- $(x, y) \rightarrow (x, 3)$
- $(x, y) \rightarrow (0, y)$
- $(x, y) \rightarrow (x, 0)$
- pour to 3 litre from 4 litre until 3-litre is full
- pour to 4 from 3-litre until 3 litre is empty.

Test Goal:

check if:
 $x = 2$ ✓
means goal achieved.

Algorithm:

- ① start from initial state $(0, 0)$:
- ② check if current state is Goal.
- ③ perform all possible steps
actions until goal is
achieved
 - Fill jug
 - empty jug
 - Pour from one into other
- ④ Add new states to search
structure (stack or queue).

(5) Repeat until goal state is reached.

(6) Print the sequence of water jugs states.

Dry Run using BFS, DFS

BFS

$(0,0)$
 $\rightarrow (4,0)$
 $\rightarrow (1,3)$
 $\rightarrow (1,0)$
 $\rightarrow (0,1)$
 $\rightarrow (4,1)$
 $\rightarrow (2,3)$
 $\rightarrow (2,0) \checkmark$

DFS.

$(0,0)$
 $\rightarrow (0,3)$
 $\rightarrow (3,0)$
 $\rightarrow (3,3)$
 $\rightarrow (4,2)$
 $\rightarrow (0,2)$
 $\rightarrow (2,0) \checkmark$